

KILLARA HIGH SCHOOL
PRELIMINARY COURSE FINAL EXAMINATION 2001
EXTENSION 1 MATHEMATICS

Time allowed : 2 Hours

- *All questions may be attempted
- *Show all necessary working
- *Board approved calculators may be used
- *START EACH QUESTION ON A NEW PAGE

Question 1

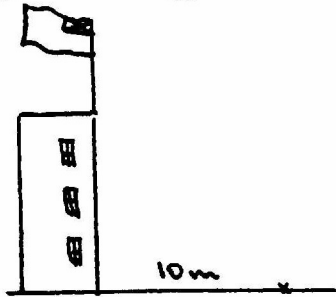
- a. Solve i. $\frac{3}{|x+1|} < 1$ ii. $\frac{3}{x+1} < 1$ (6)
- b. Given that P is the point (-3,4) and Q is (1,1), find the coordinates of the point which divides the interval PQ
- i. internally in the ratio 3:2 (2)
 - ii. externally in the ratio 3:2 .
- c. Find the angle between the lines $x - 3y + 17 = 0$ and $2x + 5y - 1 = 0$.
- d. i. For the parabola $-12y = (x - 5)^2$ find (3)
- ii. the coordinates of the focus
 - iii. the focal length
 - iv. the equation of the directrix .

Question 2

- a. Find the derivative of
- i. $(2x - 3)^8$
 - ii. $\sqrt{x^2 + 4}$ (8)
 - iii. $\frac{4x - 3}{2x + 5}$
- iv. Show that the derivative of $(x + 1)^3(x + 2)^4$ is $(7x + 10)(x + 1)^2(x + 2)^3$.
- b. Given that A is the point (-2,0), B is (4,0) and P(x,y) moves such that PA is perpendicular to PB,
- i. show that the locus of P is a circle. (5)
 - ii. Find the coordinates of the centre and the radius of the circle.

Question 3

- a. An observer standing 10 metres from the base of a building, and on the same horizontal level as the base, observes the angle of elevation of the bottom and top of a flagpole, on top of the building, to be 42° and 54° respectively. Find the height of the flagpole. (3)

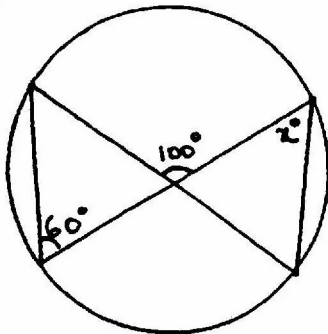


- b. Prove that $\cot\theta - \cot 2\theta = \operatorname{cosec} 2\theta$. (4)
- c. Express $4\cos\theta + 3\sin\theta$ in the form $A\sin(\theta - \theta)$ where $A > 0$. (3)
- d. Using the 't' formulae (where $t = \tan \frac{x}{2}$), solve the equation (4)
- $$\sin x + \cos x = 1 \text{ for } 0^\circ \leq x \leq 360^\circ.$$

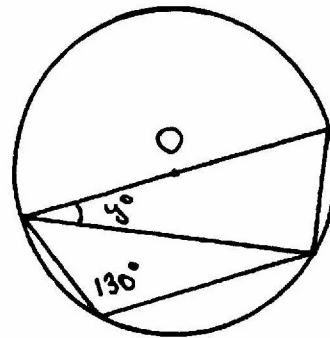
Question 4

a. Find the value of the pronumerals in each of the following diagrams.
(O is the center of the circle.)

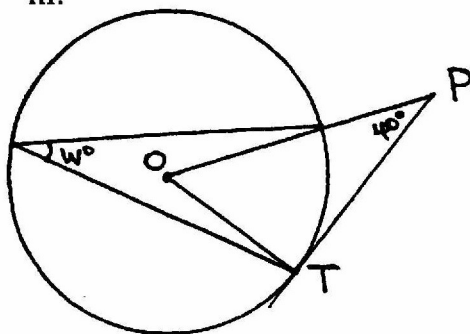
i.



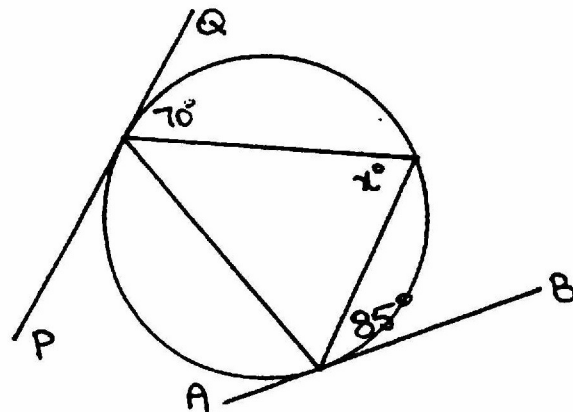
ii.



iii.



iv.

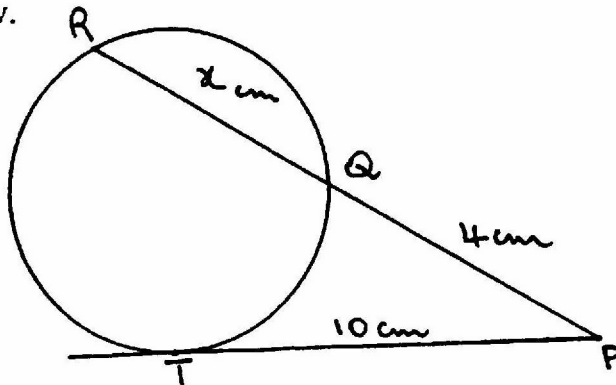


(5)

PT is a tangent.

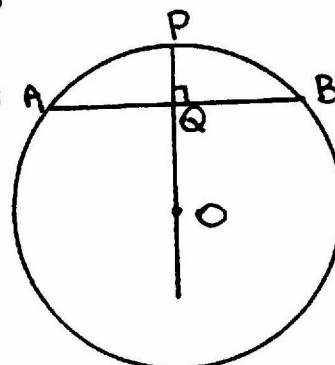
AB, PQ are tangents.

v.



PT is a tangent.

b. In the diagram, OP is perpendicular to AB. If $AB = 14$ cm and $PQ = 1$ cm, find the radius of the circle.



(2)

c. ABC is an isosceles triangle with $AB = AC$. P and Q are points on AB and AC respectively such that PQ is parallel to BC. Prove that the quadrilateral PBCQ is cyclic.

(3)

Question 5

- a. Find the quotient and remainder when $2x^3 - 4x^2 + 7x - 1$ is divided by $x - 5$. (2)
- b. When $x^3 + 3x^2 - bx - 7$ is divided by $x - 2$ the remainder is 20. Find the value of b. (2)
- c. Find all the zeros of the polynomial $P(x) = x^3 - 2x^2 - 11x + 12$. (3)
- d. Draw a neat sketch of the polynomial function $y = (x + 2)^2(x - 1)^3$. (3)
- a. If $x^3 - 8x^2 + kx - 12 = 0$ has one root equal to the sum of the other two, find the value of k. (4)

Question 6

- a. $P(2p, p^2)$ and $Q(2q, q^2)$ are points on the parabola $x^2 = 4y$.
- i. Find the equation of the chord PQ.
- ii. If PQ passes through the point $(0, -1)$, show that $pq = 1$.
- iii. Write down the equation of the chord of contact of tangents to the parabola from the point $(4, -12)$. (6)
- iv. Prove that the chord of contact of tangents drawn from any point on the line $y = -1$ is a focal chord.
- b. The tangent to the parabola at a point P on it, is perpendicular to the tangent at a point Q. If the tangents intersect at A, find the locus of A. (4)

END OF EXAMINATION

Extension 1 Mathematics - Solutions

Question 1

a. i. $\frac{3}{|x+1|} < 1$

$3 < |x+1|$ ✓

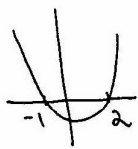
$x+1 > 3$ or $x+1 < -3$ ✓

$x > 2$ or $x < -4$ ✓

ii. $\frac{3}{x+1} \times (x+1)^2 < (x+1)^2$ ✓

$3(x+1) < (x+1)^2$

$0 < (x+1)(x-2)$ ✓



$x < -1$ or $x > 2$ ✓

b. i. $P(-3, 4)$ $Q(1, 1)$
 midpoint

$\Rightarrow \left(\frac{-3+1}{2}, \frac{4+1}{2} \right)$

ie. $\left(-\frac{1}{2}, \frac{5}{2} \right)$ ✓

ii. $P(-3, 4)$ $Q(1, 1)$
 midpoint

$\Rightarrow \left(\frac{-3+1}{2}, \frac{4+1}{2} \right)$

ie. $(-1, 2.5)$ ✓

c. $x-3y+17=0 \Rightarrow m_1 = \frac{1}{3}$ ✓
 $2x+5y-1=0 \Rightarrow m_2 = -\frac{2}{5}$

$\tan \theta = \frac{m_1 - m_2}{1 + m_1 m_2}$

$= \frac{\frac{1}{3} + \frac{2}{5}}{1 + \frac{1}{3} \times -\frac{2}{5}}$ ✓

$= \frac{\frac{11}{15}}{1 - \frac{2}{15}}$

$= \frac{11}{13}$

$\therefore \theta = 40^\circ (14)$ ✓

d. i. $(5, 1)$ ✓

ii. $a = 3$ ✓

iii. $y = 2$ ✓

Question 2

a. i. $8(2x-3)^7 \times 2$
 $= 16(2x-3)^7$

ii. $\frac{1}{2}(x^2+4)^{-\frac{1}{2}} \times 2x$
 $= x(x^2+4)^{-\frac{1}{2}}$

iii. $\frac{(2x+5)(4) - (4x-3)(2)}{(2x+5)^2}$ ✓

$= \frac{26}{(2x+5)^2}$ ✓

iv. $(x+1)^3 \times 4(x+2)^3 + (x+2)^4 \times 3(x+1)^2$

$= (x+1)^2(x+2)^3 [4(x+1) + 3(x+2)]$ ✓

$= (x+1)^2(x+2)^3 (7x+10)$

b. i. $m_{PA} = \frac{y}{x+2}$, $m_{PB} = \frac{y}{x-4}$ ✓

$m_{PA} \times m_{PB} = -1$

$\frac{y^2}{(x+2)(x-4)} = -1$ ✓

$y^2 = -(x^2 - 2x - 8)$

$y^2 = -x^2 + 2x + 8$

$x^2 + y^2 - 2x - 8 = 0$ ✓

This is general equation of a circle.

ii. $(x^2 - 2x + 1) + y^2 = 8 + 1$
 $(x-1)^2 + y^2 = 9$
 Centre is $(1, 0)$, $R = 3$ ✓

Question 3



$$\begin{cases} \tan 42^\circ = \frac{y}{10} \Rightarrow y = 10 \tan 42^\circ \\ \tan 54^\circ = \frac{x+y}{10} \end{cases}$$

$$\begin{aligned} \Rightarrow x+y &= 10 \tan 54^\circ \\ x+10 \tan 42^\circ &= 10 \tan 54^\circ \quad \checkmark \\ x &= 4.8 \text{ m} \quad (4.7597 \dots) \end{aligned}$$

$$\begin{aligned} \text{b. LHS} &= \frac{\cos \theta}{\sin \theta} - \frac{\cos 2\theta}{\sin 2\theta} \quad \checkmark \\ &= \frac{\cos \theta}{\sin \theta} - \frac{2\cos^2 \theta - 1}{2\sin \theta \cos \theta} \quad \checkmark \\ &= \frac{2\cos^2 \theta - (2\cos^2 \theta - 1)}{2\sin \theta \cos \theta} \\ &= \frac{1}{2\sin \theta \cos \theta} \quad \checkmark \\ &= \csc 2\theta \quad \checkmark \\ &= \text{RHS} \end{aligned}$$

$$\begin{aligned} \text{c. } 4\cos \theta + 3\sin \theta \\ &= A \sin \theta \cos \alpha - A \cos \theta \sin \alpha \quad \checkmark \end{aligned}$$

$$A \sin \alpha = -4 \quad \text{--- (1)} \quad \checkmark$$

$$A \cos \alpha = 3 \quad \text{--- (2)} \quad \checkmark$$

$$\text{(1)}^2 + \text{(2)}^2 \Rightarrow A = 5 \quad \checkmark$$

$$\begin{aligned} \text{(1)} + \text{(2)} \Rightarrow \tan \alpha &= -\frac{4}{3} \\ \alpha &= 307^\circ \quad (306^\circ 52') \quad \checkmark \end{aligned}$$

$$\begin{aligned} \text{d. } \sin x + \cos x &= 1 \\ \frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2} &= 1 \quad \checkmark \\ 2t+1-t^2 &= 1+t^2 \\ 2t^2-2t &= 0 \\ 2t(t-1) &= 0 \\ t=0 \quad \text{or} \quad t=1 \quad \checkmark \end{aligned}$$

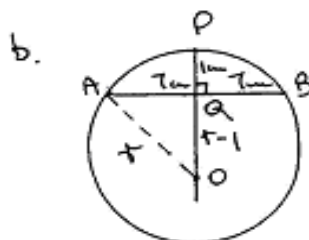
$$\begin{aligned} \tan \frac{x}{2} &= 0 \\ \frac{x}{2} &= 0^\circ, 180^\circ, 360^\circ \\ x &= 0^\circ, 360^\circ \quad \checkmark \end{aligned}$$

$$\begin{aligned} \tan \frac{x}{2} &= 1 \\ \frac{x}{2} &= 45^\circ, 225^\circ \\ x &= 90^\circ, \quad \checkmark \end{aligned}$$

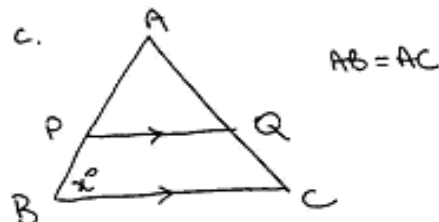
$$\text{ie. } x = 0^\circ, 90^\circ, 360^\circ$$

Question 4

- i. $x = 40 \quad \checkmark$
- ii. $y = 40 \quad \checkmark$
- iii. $w = 25 \quad \checkmark$
- iv. $x = 25 \quad \checkmark$
- v. $x = 5 \quad \checkmark$



$$\begin{aligned} r^2 &= (t-1)^2 + 7^2 \quad \checkmark \\ t^2 &= t^2 - 2t + 1 + 49 \\ 2t &= 50 \\ t &= 25 \quad \checkmark \end{aligned}$$



$$\begin{aligned} \angle ACB &= \angle ABC = x^\circ \quad (AB = AC) \quad \checkmark \\ \angle AQP &= \angle ACB = x^\circ \quad (\text{corr. } \angle\text{'s} \\ &\quad \text{+ } PQ \parallel BC) \quad \checkmark \end{aligned}$$

$$\begin{aligned} \text{ie. } \angle AQP &= \angle PBC \\ \therefore PBCQ &\text{ is cyclic} \\ \text{Ext. } \angle \text{ of quad.} &= \text{int. opp. } \angle \quad \checkmark \end{aligned}$$

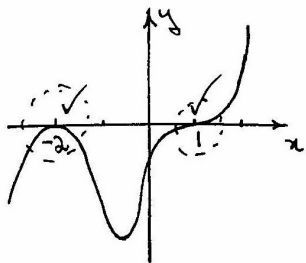
Question 5

a.
$$\begin{array}{r} 2x^3 + 6x + 37 \\ x-5 \overline{) 2x^3 - 4x^2 + 7x - 1} \\ \underline{2x^3 - 10x^2} \\ 6x^2 + 7x - 1 \\ \underline{6x^2 - 30x} \\ 37x - 1 \\ \underline{37x - 185} \\ 184 \end{array}$$

b. $P(x) = x^3 + 3x^2 - bx - 7$
 $P(2) = 8 + 12 - 2b - 7 = 13 - 2b$
 $\therefore 13 - 2b = 20$
 $b = -\frac{7}{2}$

c. $P(x) = x^3 - 2x^2 - 11x + 12$
 $P(1) = 0, P(-3) = 0, P(4) = 0$
 Zeros are $1, -3, 4$

d.



e. Let roots be $\alpha, \beta, \alpha + \beta$
 $\alpha + \beta + (\alpha + \beta) = 8 \Rightarrow \alpha + \beta = 4 \dots (1)$
 $\alpha\beta + \alpha(\alpha + \beta) + \beta(\alpha + \beta) = k$
 $\Rightarrow \alpha\beta + (\alpha + \beta)^2 = k \dots (2)$
 $\alpha\beta(\alpha + \beta) = 12 \dots (3)$
 Subst (1) into (3)
 $\Rightarrow \alpha\beta = 3 \dots (4)$
 Subst. (1) & (4) into (2)
 $\Rightarrow 3 + 4^2 = k$
 $k = 19$

Let roots be α, β, γ & $\alpha = \beta + \gamma$
 $\alpha + \beta + \gamma = 8$
 $2\alpha = 8$
 $\alpha = 4$ is one root
 etc.

Question 6

a. i. $M_{PQ} = \frac{p^2 - q^2}{2p - 2q}$
 $= \frac{p+q}{2}$

Equation of PQ is
 $y - p^2 = \frac{p+q}{2}(x - 2p)$
 $y - p^2 = \frac{1}{2}(p+q)x - p^2 - pq$
 i.e. $y = \frac{1}{2}(p+q)x - pq$

ii. If PQ passes thro' (0, -1)
 $-1 = \frac{1}{2}(p+q)(0) - pq$
 i.e. $pq = 1$

iii. $xx_0 = 2a(y + y_0)$
 $4x = 2(y - 1)$
 (or $2x - y + 1 = 0, y = 2x + 1$)

iv. Let $(x_1, -1)$ be any pt.
 on $y = -1$.
 Chord of contact is
 $xx = 2(y - 1)$

Focus is (0, 1).
 Subst. coords of focus
 $\Rightarrow 0x = 2(1 - 1)$
 $\therefore$ Focal chord.

b. $y = \frac{x^2}{4a} \Rightarrow y' = \frac{x}{2a}$

At $P(2ap, ap^2)$ $y' = p$
 Tangent at P is
 $y - ap^2 = p(x - 2ap)$
 i.e. $y = px - ap^2 \dots (1)$

Similarly, tangent at Q is
 $y = qx - aq^2 \dots (2)$

For pt. of intersection
 $(1) \times q \Rightarrow qy = pqx - apq^2 \dots (3)$

$(2) \times p \Rightarrow py = pqx - apq^2 \dots (4)$

$(3) - (4) \Rightarrow (p - q)y = -apq(p - q)$
 i.e. $y = +apq$

[x is a (p+q)]

Since tangents at P & Q are perpendicular

$pq = -1$

$\therefore y = +ax - 1$
 i.e. $y = a$

is the equation of the locus.